

The Arrow's Alibi: Preferred-Frame Safety and Quantum Coherence from Sparse Becoming

Allen Proxmire

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Abstract

Event Density (ED) is a discrete, relational substrate whose one special feature is that becoming is an irreversible *commitment* — the arrow of time written into the law rather than the boundary conditions. The gravity line (GR-I/II, KM-I/II) shows this substrate produces an emergent Einstein weak field and a gravitational class that is **khronometric** — Einstein's tensor sector plus one propagating scalar, the *khronon*, the arrow made dynamical. Khronometric theories carry a notorious liability: a preferred frame generically produces post-Newtonian preferred-frame effects (the parameters α_1, α_2) of order unity, in gross conflict with the experimental bound $|\alpha_1| < 2.5 \times 10^{-5}$. This paper shows ED evades that fate, and *why* — by a mechanism internal to its ontology rather than a tuning.

The second preferred-frame parameter, α_2 , vanishes *exactly* — a structural consequence of ED's two gravitational cones both being at light speed ($c_T = c$, $c_s = c$), independent of any tuning, so the tighter of the two experimental bounds is satisfied for free. The remaining parameter reduces to the *local* khronon stiffness, $\alpha_1 = -4 \lambda_{\text{local}}$. We derive that $\lambda_{\text{local}} \sim \rho_{\text{event}}/\rho_{\text{Planck}}$ — the ratio of the local density of committed events to the Planck density — because the metric stiffness comes from the *always-on* reciprocal sharing of the substrate while the khronon stiffness comes from the *sparse* act of commitment. The decisive step is that commitment **must** be sparse: dense commitment (a determination at every fundamental tick) is exactly the quantum Zeno limit, which freezes all evolution; a universe measured continuously is a universe that never evolves, and so has no quantum mechanics. ED has a quantum sector, so the sparse branch is *forced*, not chosen. Consequently $\lambda_{\text{local}} \ll 1$ in every sub-Planck environment, and α_1 is suppressed below the bound by $\gtrsim 70$ orders of magnitude. The galactic modified-dynamics scale $a_0 = cH_0/(2\pi)$ is untouched: it arises from the khronon's *cosmological* face — a dipole projection of the cosmic decoupling surface $R_H = c/H_0$ — not from the local stiffness, and is in fact strongest exactly where the local stiffness is smallest.

The result has a unifying reading. **The same sparseness that lets the universe remain coherent is what keeps relativity intact.** Sparse becoming gives ED quantum coherence (evolution between rare commitments), gravitational time dilation (a position-dependent rate of becoming — a uniform tick cannot bend), and preferred-frame safety (a tiny local khronon coupling) — *one structure, three faces*. ED is therefore a **screened khronometric theory**: Einstein locally, Milgrom cosmologically, distinct from General Relativity only at the Planck-density frontier. Preferred-frame safety is not a constraint ED satisfies; it is a consequence of

ED’s being a quantum, time-dilating theory at all.

What this paper does not claim

To keep the program’s epistemic discipline explicit, the following are *not* claimed here:

- that ED derives the full MOND interpolation function (only its asymptotic limits are fixed; KM-I, Paper 036);
 - that ED excludes particle dark matter (the cosmological dust sector is *addressed*, *not discharged*; KM-II);
 - that ED derives H_0 or cosmology (H_0 is value-inherited from observation);
 - that $\alpha_1 = 0$ exactly — only that it is suppressed by $\gtrsim 70$ orders of magnitude below the bound;
 - that “sparse commitment” is a new primitive — it is *derived* from the distinction $P11 \neq P13$ together with ED’s quantum sector.
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1. Introduction

1.1 The kill-switch

The ED gravity line is built in stages. A discrete substrate with a bandwidth field b produces a Lorentzian metric and the Einstein weak field, including factor-of-two light bending (GR-I); a uniqueness argument plus a mode count shows the gravitational class is khronometric — Einstein’s two tensor modes plus one scalar, the khronon (GR-II); the khronon’s deep-infrared regime reproduces MOND phenomenology with the transition acceleration $a_0 = cH_0/(2\pi)$ (KM-I, building on Papers 028–031); and its cosmological face carries a dark-fluid sector and the cosmological-constant scale (KM-II). The capstone One-Field letter gathers these into a single claim — *one field, one scale, five roles* — and, in the discipline that defines this program, names the three ways the theory can die. The first kill-switch is the **preferred-frame parameters** α_1, α_2 : “uncomputed, must pass solar-system and pulsar bounds.”

This is not a small liability. Preferred-frame theories of gravity are *easy to kill*. A preferred rest frame generically makes a gravitating system’s dynamics depend on its velocity through that frame, an effect parametrized in the post-Newtonian formalism by α_1 (linear in the velocity) and α_2 (quadratic). These are among the most tightly constrained quantities in gravitation — bounded by lunar laser ranging and pulsar timing to roughly $|\alpha_1| \lesssim 10^{-4}$ and $|\alpha_2| \lesssim 10^{-7}$ — and the generic expectation for a theory that breaks Lorentz invariance at the fundamental scale is that they come out of order unity. A theory that predicts $\alpha_1 = O(1)$ is excluded by a factor of 10^4 or more. ED, having put the arrow of time into the law, has a preferred frame by construction. So the question is sharp: does the arrow leave a preferred-frame footprint large enough to kill the theory?

1.2 What this paper claims

It does not. **The arrow is real, but it leaves no shadow.** This paper computes the preferred-frame coupling from ED’s primitives and finds it suppressed below the experimental bound by

tens of orders of magnitude — not by adjusting a parameter, but as a forced consequence of what ED is.

The argument has a clean spine, and the spine is the surprise. The preferred-frame parameter α_1 probes the *local* stiffness of the preferred foliation, and that stiffness is set by the local density of committed events relative to the Planck density. That density is small in vacuum because **the universe becomes — but not all at once**. And it *must* become sparsely, because dense becoming — a determination at every fundamental tick — is the quantum Zeno limit, which freezes evolution entirely. **Dense becoming would freeze the universe; sparse becoming lets it evolve**. ED has quantum mechanics, so it lives on the sparse branch, and on the sparse branch the preferred-frame coupling is tiny.

The reader should therefore hold, from the start, the unification that organizes the paper:

Quantum mechanics and relativistic time dilation emerge from the same structural fact: becoming is rare. Persistent coherence (quantum mechanics), a position-dependent clock (gravitational time dilation), and a faint preferred-frame footprint (α_1 -safety) are three faces of one structure — sparse commitment. **Preferred-frame safety is not a constraint ED satisfies — it is a consequence of ED’s ontology.**

ED’s arrow does not pick out a preferred frame so much as a preferred *time*: it picks out *when* reality becomes, not *where*. That distinction — temporal asymmetry versus spatial preference — is the technical heart of why the footprint is faint.

One structure, three faces: sparse becoming $\rightarrow$ quantum coherence; sparse becoming $\rightarrow$ gravitational time dilation; sparse becoming $\rightarrow$ preferred-frame silence.

Result in one sentence: ED is a screened khronometric theory whose preferred-frame coupling is suppressed by the very same sparse-becoming structure that gives it quantum coherence.

1.3 How the result was reached (a note on method)

This conclusion was not assumed. It was reached by genuinely trying to make α_1 fail. A standard effective-field-theory estimate — Lorentz violation generated at the Planck scale yields $O(1)$ low-energy couplings — was applied first, and on that estimate ED was in flat tension with the bound; the front looked fatal. The favorable result emerged only when the generic estimate was discarded in favor of ED’s actual structure, which the estimate had silently assumed away: that the substrate’s “background” is not a uniform Planck-density medium but a *non-uniform density of becoming*, and that the symmetry-breaking events are not every tick but the rare, decoherence-like acts of commitment. The result is offered with that history attached, because a theory that states precisely where it nearly died is harder to dismiss than one that reports only its survivals.

Reader map.

- §2 — the two faces of the khronon (local stiffness vs. cosmic boundary);
- §3 — the reduction $\alpha_1 = -4\lambda_{\text{local}}$;
- §4 — the derivation $\lambda_{\text{local}} \sim \rho_{\text{event}}/\rho_{\text{Planck}}$;
- §5 — why commitment is *forced* sparse;
- §6 — the magnitude ($\gtrsim 70$ -order safety);

- §7 — the MOND reconciliation;
 - §8 — the unification (one structure, three faces);
 - §9–§11 — relation to the arc, falsifiers, and the position statement.
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2. Background: ED’s Two Faces of the Khronon

The single most important distinction in this paper is that the khronon enters ED’s gravity in two structurally different ways, at two vastly separated scales. They were historically easy to conflate; separating them is what dissolves the apparent conflict between preferred-frame safety and the existence of a galactic modified-dynamics scale.

2.1 The local face: foliation stiffness λ_{local}

GR-II establishes that ED’s gravitational class is khronometric: the arrow (commitment irreversibility, the homogeneous tick) singles out a preferred time-foliation at the level of the dynamics, reducing the gauge group from full diffeomorphism invariance to foliation-preserving diffeomorphisms, and the propagating-mode count becomes two tensor polarizations plus one scalar — the khronon. The khronon is the *preferred-foliation field*, the arrow of time made dynamical.

Locally, the khronon has a kinetic weight — a stiffness with which the foliation resists deformation at a point. In the khronometric action this is the coefficient λ of the $(\nabla \cdot u)^2$ (foliation-expansion) term, where u_μ is the unit normal to the foliation. We write it λ_{local} to mark that it is a *local* property of the field. As §3 makes precise, the post-Newtonian preferred-frame parameter probes exactly this local stiffness. The local face is the one experiments in the Solar System constrain.

2.2 The cosmological face: the cosmic decoupling surface

The same foliation has a cosmological face — the background against which the whole substrate becomes. Paper 028 identifies the **cosmic decoupling surface** at radius

$$R_H = \frac{c}{H_0},$$

the substrate-level boundary beyond which kernel-mediated influence (propagating at the substrate speed c) is outrun by Hubble recession (at rate $H_0 R$). This surface is not a local coupling; it is a single cosmic-scale structure set by H_0 , the Hubble flow that *is* the khronon’s background rest frame. Paper 029 shows that, for an accelerating chain, the projection of this surface’s anisotropic content onto the chain’s residual $SO(2)$ symmetry produces a characteristic acceleration $a_0 = cH_0/(2\pi)$ — the MOND transition scale. The cosmological face is the one galactic dynamics probes.

2.3 Why they were conflated, and why they must be separated

Both effects involve “the khronon,” and both are “preferred-frame” in the loose sense that both know about the cosmic rest frame. It is therefore natural — and, in the analysis that led to this paper, was at first assumed — that a theory’s galactic modified-dynamics signal

and its post-Newtonian preferred-frame parameters are governed by *the same coupling*, so that suppressing one suppresses the other. They are not. The local stiffness λ_{local} is a property of the foliation *at a point*, set by the local density of becoming; the cosmic acceleration $a_0 = cH_0/(2\pi)$ is a property of the cosmic *boundary*, set by H_0 . One scales with the local event density; the other is fixed by the cosmic rate. **A local whisper and a cosmic echo are different sounds.** Keeping them separate is the whole reconciliation (§7); the rest of the paper computes the local face and shows it is tiny.

3. The PPN Reduction: $\alpha_1 = -4\lambda_{\text{local}}$

The post-Newtonian preferred-frame parameters of a khronometric theory are known functions of its couplings. In the standard parametrization (three dimensionless couplings α , β , λ , with α_1, α_2 both proportional to the combination $\alpha - 2\beta$),

$$\alpha_1 = \frac{4(\alpha - 2\beta)}{\beta - 1}.$$

ED’s established structural results fix two of the three couplings. The single shared causal cone (GR-II, one transport process P05) makes the tensor speed equal the light speed, $c_T = c$, which in this parametrization is $\beta = 0$. The khronon propagates at the speed of light in vacuum, $c_s = c$ (the irreversible commitment-reserve sector is dissipative, not kinetic; GR-III), which with $\beta = 0$ ties the remaining couplings through the scalar-speed relation, giving $\alpha = \lambda/(1 + 2\lambda)$ where λ is the khronon’s expansion coupling. With $\beta = 0$, the preferred-frame parameter collapses to a single number:

$$\boxed{\alpha_1 = -4\alpha = -4\lambda_{\text{local}}} \quad (\beta = 0),$$

where, for the small couplings of interest, $\alpha = \lambda/(1 + 2\lambda) \approx \lambda \equiv \lambda_{\text{local}}$. (*Structural — the khronometric PPN formula evaluated on ED’s derived constraints $c_T = c$, $c_s = c$.*)

And α_2 vanishes outright. ED’s two luminal constraints do more than reduce α_1 to one coupling — they kill α_2 identically. Besides the shared $(\alpha - 2\beta)$ factor, the khronometric α_2 carries a second factor that vanishes precisely on the scalar-luminal surface $\lambda = \alpha/(1 - 2\alpha)$: at $\beta = 0$ the published α_2 numerator is $-\lambda + \alpha + 2\alpha\lambda$, which is *exactly zero* once $c_s = c$ is imposed. So

$$\boxed{\alpha_2 = 0} \quad (c_T = c, \ c_s = c),$$

not merely suppressed but exactly zero, for a reason **independent of the sparsity argument** below — a general feature of any khronometric theory with both cones at light speed. (Verified against the published khronometric PPN formulas — Hořava-gravity arXiv:1105.5149 eq. 31, Blas-Sibiryakov arXiv:1412.4828 — and checked numerically.) Consequently the *tighter* of the two preferred-frame bounds, $|\alpha_2| \lesssim 10^{-7}$, is satisfied structurally, and **only α_1 carries the falsification number**, against the looser bound $|\alpha_1| \lesssim 10^{-4}$. The rest of this paper computes α_1 . (*Derived + literature-verified.*)

The reduction makes the target precise: α_1 **probes the local khronon stiffness, and nothing else**. It does not probe the cosmic boundary, the matter content, or the global foliation; it is set entirely by how stiffly the preferred foliation resists deformation *here*. Two corollaries matter. First, $\lambda_{\text{local}} = 0$ is not available: it would force $\lambda = 0$ and hence all khronometric couplings to zero, which is pure General Relativity with no propagating khronon — and ED has a propagating khronon. So α_1 is genuinely nonzero. Second, the whole question of preferred-frame safety is now the single question: *how large is λ_{local} ?* The remaining sections answer it from the primitives.

4. Deriving λ_{local} from ED’s Primitives

The decisive structural fact is that the two stiffnesses entering $\lambda_{\text{local}} = f^2/M_P^2$ — the khronon stiffness f^2 and the gravitational (metric) stiffness M_P^2 — come from *different primitives* with *different* dependence on the local state of the substrate.

4.1 The metric stiffness is always on (P02)

The gravitational stiffness M_P^2 is the coefficient of the spatial-curvature energy — the rigidity of the metric field b . In ED the metric band is a **reciprocal, shared** edge record (P02): the connection strength between two loci is a single quantity belonging to both, and it exists and equilibrates across adjacency at *every* link, *every* tick (P13), whether or not anything commits. The metric is the substrate’s *standing structure*. Its stiffness is therefore set by the always-present substrate connectivity — Planck-dense and **independent of the local density of events**:

$$M_P^2 \sim \frac{s_{02}}{\ell_P^2},$$

with s_{02} the dimensionless per-tick sharing fraction (a P02/P04 band quantity). This is why gravity is full-strength everywhere, including in near-empty space.

4.2 The khronon stiffness tracks commitment (P11)

The khronon stiffness f^2 is the coefficient of the foliation-deformation energy. But the foliation is the structure of *constant commitment-count*: it is pinned only **where commitments fire** (P11). Between commitments the foliation carries no constraint — there is nothing holding “now” in place. So f^2 is set by the **density of commitment-pinning events**, which is a property of the local state, not the standing connectivity:

$$f^2 \sim k_{11} \rho_{\text{event}} \ell_P^2,$$

with k_{11} the dimensionless commitment factor and ρ_{event} the local density of committed (irreversible determination) events per unit four-volume.

4.3 Their ratio is the commitment sparsity

Form the ratio. Define the dimensionless **commitment sparsity** $s \equiv \rho_{\text{event}}/\rho_{\text{Planck}}$, where $\rho_{\text{Planck}} = \ell_P^{-4}$ is the maximal density (every Planck cell committing every tick), so $0 \leq s \leq 1$. The Planck scales cancel:

$$\lambda_{\text{local}} = \frac{f^2}{M_P^2} = \frac{k_{11}}{s_{02}} \frac{\rho_{\text{event}}}{\rho_{\text{Planck}}} \sim \frac{\rho_{\text{event}}}{\rho_{\text{Planck}}}$$

with the prefactor k_{11}/s_{02} of order unity — it is the same band-fraction ratio that fixes the gravitational coupling, $\kappa/D = 8\pi G$, in the dynamical rule (GR-III). (*Derived from the primitive origins of the two stiffnesses; the linear scaling holds in the dilute regime where commitments do not interact, which is the vacuum regime relevant to α_1 . Near Planck density the dilute approximation fails and λ_{local} saturates toward order unity.*)

This is the crux. The symmetry-breaking scale of the khronon is **not** the Planck scale; it is the local event-density scale, $f^2 \sim \rho_{\text{event}} \ell_P^2$. The standard “Planck-scale breaking $\Rightarrow O(1)$ couplings” estimate assumed a uniform Planck-dense background. ED has no such background. **Gravity (sharing) is everywhere; the preferred frame (commitment) is only where events become.**

4.4 The event-density hierarchy

Because ρ_{event} tracks where determination actually happens, it is profoundly non-uniform:

$$\rho_{\text{event}}^{\text{matter}} \gg \rho_{\text{event}}^{\text{vacuum}} \gg \rho_{\text{event}}^{\text{cosmological floor}}.$$

Dense near matter (where persistent interactions force determination), sparse in vacuum (where coherent, undetermined states persist), and bounded below by the residual cosmic rate. **In ED, gravity is literally the density of committed events** — so this hierarchy, *the density of becoming*, is the same structure that sources gravity, and it is what makes λ_{local} an environment-dependent quantity rather than a constant. The next section shows the hierarchy’s low end is *forced* to be far below Planck.

5. The Sparse vs. Dense Fork

Everything now turns on one question: is commitment (P11) a *sparse* physical determination event, or a *dense* process that fires at every fundamental tick (P13)? On the sparse branch $\lambda_{\text{local}} \ll 1$ and α_1 is safe; on the dense branch $\rho_{\text{event}} = \rho_{\text{Planck}}$, $\lambda_{\text{local}} = O(1)$, and $\alpha_1 = O(1)$ — fatal. We show the sparse branch is forced.

5.1 P11 acts on indeterminacy; P13 is the tick

The two primitives are different in kind. **P13** is the tick — a homogeneous, fundamental rhythm of becoming, the substrate’s intrinsic *clock*. It is universal: it sets the maximal *rate* at which determination can proceed. **P11** is commitment — the irreversible act by which an *indeterminate* degree of freedom becomes a *determinate* fact. It is an *event* that acts on indeterminacy: the substrate’s analogue of decoherence or measurement, “superposition $\rightarrow$ definite outcome” being precisely “indeterminate $\rightarrow$ determinate.” In one line: **P13 is the rate at which becoming can proceed; P11 is when becoming does proceed.** The dense branch asserts the two are the same — that every tick *is* a determination. The sparse branch asserts that commitment

fires only where there is indeterminacy to resolve. These cannot both be primitive without one collapsing into the other.

5.2 Dense commitment has nothing to commit

P11 converts indeterminate degrees of freedom into determinate ones. For commitment to be *ongoing*, there must be *persistent* indeterminacy. If P11 fired on everything at every tick, then after one tick the substrate would be fully determinate and commitment would simply stop — there would be nothing left to commit. **If everything became at once, nothing could change.** A universe that commits every tick is a universe that has already finished becoming, and then never moves. Ongoing becoming therefore *requires* persistent indeterminacy, which is exactly the statement that commitment is sparse relative to the substrate.

5.3 V1 forbids a global collapse

The arrow gives a *direction* of time. A direction does not, by itself, give a preferred *foliation* (a global “now”): for that, the determination of “now” would have to be synchronized across space. Finite-speed causation (V1, a finite-width retarded kernel; the single cone P05) forbids exactly this synchronization at the fundamental level — spacelike-separated regions are causally disconnected, so their commitments are independent. There is no instantaneous global collapse. Commitment is local, and the cosmic rest frame is selected *spontaneously* by the matter distribution (the Hubble flow), as the cosmic microwave background frame is selected — not by an explicit global slicing. A dense, globally-synchronized every-tick collapse is what V1 most directly forbids.

5.4 Gravity itself requires non-uniform becoming

ED’s gravity *is* event density, and gravitational fields are structured: strong near matter, weak in vacuum. Were commitment uniform and Planck-dense everywhere, the event density would be uniform, and gravity would carry no information about matter — it would be a structureless constant rather than a field. The matter-sourced gravitational field that ED actually has *requires* the density of becoming to track determination, not the clock. The density-of-becoming ontology is coherent only on the sparse branch.

And the substrate’s own clock confirms it. **Time dilates because becoming is uneven.** Gravitational time dilation requires the proper tick rate to vary from place to place — the lapse N (with $N^2 \sim b$) must be position-dependent. A uniform, every-tick, identical becoming would tick the same everywhere, and there would be no lapse variation, no redshift, no time dilation. **A uniform tick cannot bend; a bending tick is what we call gravity.** That the clock bends at all is a sign that becoming is non-uniform.

5.5 The quantum Zeno forcing

The previous three arguments make sparse commitment necessary for ongoing change, for causal structure, and for gravity. The fourth makes it *unconditional*, and it is the decisive one.

ED has a quantum sector: superposition, interference, the U(1) phase (P09). A superposition is *persistent indeterminacy* — a degree of freedom held coherently uncommitted across many ticks. Now suppose commitment were dense: a determination at every tick. Then every coherent superposition would be measured every Planck time. This is precisely the **quantum Zeno limit**:

continuous measurement does not merely degrade interference, it *freezes evolution entirely* — a continuously-observed system is pinned to its instantaneous eigenstate and cannot change. **Dense commitment is the quantum Zeno limit: a universe measured continuously is a universe that never evolves.** It has no superposition, no interference, no coherence, no unitary evolution — and, by Zeno, no dynamics at all.

Therefore the dense branch is not merely disfavored; it is *incompatible with ED having quantum mechanics*. Since ED has a quantum sector, commitment cannot be dense. The vacuum, in particular, is coherent — determination there is rare — so the vacuum commitment density is far below Planck.

5.6 Verdict: commitment is sparse, and it is forced

P11 $\neq$ *P13*. **Commitment is a sparse, local, decoherence-like determination event — the tick is the universal clock, commitment is what happens within it where there is indeterminacy to resolve.** The dense branch is excluded decisively, because it is the Zeno limit and would abolish ED’s quantum mechanics. **Sparse commitment is not a choice; it is the condition for there being a world at all.** (*Structural — forced by ED’s quantum sector, with three independent supporting forcings: P11’s logic, V1 causal structure, and the density-of-becoming origin of gravity.*)

Hence $\rho_{\text{event}} \ll \rho_{\text{Planck}}$ in the vacuum, and $\lambda_{\text{local}} \ll 1$.

6. Magnitude: α_1 Safe by $\gtrsim 70$ Orders

With sparseness forced, the magnitude follows. On the sparse branch ρ_{event} tracks the local physical activity, and every relevant density is fantastically sub-Planck ($\rho_{\text{Planck}} \approx 5 \times 10^{113} \text{ J/m}^3$).

6.1 Event-density estimates

Regime	candidate local density	$\rho_{\text{event}}/\rho_{\text{Planck}}$
Solar-System vacuum (1 AU), field energy $g^2/8\pi G$	$\approx 2 \times 10^4 \text{ J/m}^3$	$\approx 10^{-110}$
1 AU, ambient matter (solar wind)	$\approx 10^{-3} \text{ J/m}^3$	$\approx 10^{-116}$
source (mean solar density, ρc^2)	$\approx 10^{20} \text{ J/m}^3$	$\approx 10^{-93}$
cosmological floor (ρ_{Λ})	—	$\approx 10^{-122}$
(extreme stress-test: nuclear density)	$\approx 10^{35} \text{ J/m}^3$	$\approx 10^{-78}$

6.2–6.4 From λ_{local} to α_1 , and the bound

Taking the most conservative (densest) plausible identification of the density the khronon samples — the source density itself —

$$\lambda_{\text{local}} \lesssim 10^{-93}, \quad \alpha_1 = -4 \lambda_{\text{local}} \lesssim 4 \times 10^{-93},$$

against the experimental bound $|\alpha_1| < 2.5 \times 10^{-5}$. The margin is $\gtrsim 88$ orders of magnitude, and even the absurd nuclear-density stress-test clears the bound by $\gtrsim 70$. **The margin is so vast that the residual uncertainty in which density the khronon samples is irrelevant to the verdict.** (*The mechanism — density-suppression — is derived; the numerical value is an order-of-magnitude estimate, robust by $\gtrsim 70$ orders across all identifications. No precise value is asserted; the claim is the suppression, not a digit.*)

6.5 ED is observationally General Relativity

A corollary worth stating plainly: λ_{local} reaches order unity only at $\rho_{\text{event}} \sim \rho_{\text{Planck}}$ — i.e. at *Planck density* (black-hole interiors, the very early universe), not at ordinary matter. For all ordinary densities — Solar System, stars, even neutron stars — $\lambda_{\text{local}} \ll 1$, so the khronometric deviations from General Relativity are unobservably small. **ED is observationally General Relativity everywhere gravity has been tested, and khronometrically distinct only at the Planck-density frontier** — which is where quantum gravity is expected to matter in any case. **The arrow is real, but its footprint is faint — suppressed by the same coherence that makes quantum mechanics possible.**

7. MOND Reconciliation

A natural worry: if the local khronon coupling is tiny everywhere, how does ED’s galactic modified-dynamics sector (KM-I) survive? The answer is that MOND was never a local-coupling effect. **MOND is not a local whisper of the arrow; it is the echo of the cosmic boundary.**

7.1 MOND does not use λ_{local}

The MOND transition acceleration in ED is derived (Paper 029) with no reference to the local foliation stiffness. Its source is the *cosmological* face of the khronon — the cosmic decoupling surface.

7.2–7.3 a_0 from the cosmic decoupling surface

The cosmic decoupling surface sits at $R_H = c/H_0$, where kernel propagation (c) is outrun by Hubble recession ($H_0 R$). An accelerating chain (acceleration $\vec{a}$) has anisotropic substrate-graph adjacency — forward channels (along $\vec{a}$) encountered at an increased rate, backward at a decreased rate — which breaks the full rotational symmetry $SO(3)$ to a residual $SO(2)$ about the acceleration axis. The leading anisotropic mode is the dipole ($\ell = 1$); projecting the cosmic content (of scale $\rho_0 = cH_0$) onto the chain’s $|n| = 1$ response, with the azimuthal-Fourier normalization $\frac{1}{2\pi} \int_0^{2\pi} e^{-in\phi} e^{im\phi} d\phi = \delta_{nm}$ on the residual $SO(2)$, yields

$$a_0 = \frac{cH_0}{2\pi} \approx 1.08 \times 10^{-10} \text{ m/s}^2,$$

a parameter-free $\sim 10\%$ match to the measured MOND scale. This is a cosmic-boundary *kinematic* effect, fixed by H_0 . The local stiffness λ_{local} appears nowhere in it.

7.4 MOND is strongest where λ_{local} is smallest

The clincher is the direction of the scalings. MOND dominates in the low-acceleration, low-density outskirts of galaxies — exactly where the event density, and hence λ_{local} , is *smallest*. A density-tracking local coupling runs the opposite way to the MOND phenomenology. So MOND could not have been a λ_{local} effect even in principle, and Paper 029 confirms it is not: it is governed by the source acceleration relative to a_0 (an acceleration-regime statement), not by the local density.

7.5 The two faces coexist without conflict

$\alpha_1 = -4\lambda_{\text{local}}$ is set by the *local* event density and goes to zero in vacuum; $a_0 = cH_0/(2\pi)$ is set by the *cosmic* rate H_0 and is constant across a galaxy. They are different quantities at scales separated by tens of orders of magnitude; suppressing the local coupling leaves the cosmic acceleration untouched. ED therefore has MOND *and* preferred-frame safety simultaneously — one foliation field playing a local and a cosmological role. (*Structural — grounded in Paper 029’s established derivation, which uses the cosmic boundary, not the local stiffness.*)

8. Unified Interpretation: One Structure, Three Faces

The cleanest reading of the result is not “ED dodged a bound.” It is that the dodge was never optional — preferred-frame safety falls out of the same structural fact that makes ED a quantum, time-dilating theory at all. **One structure, three faces: coherence, dilation, and silence.**

ED’s dynamics, on the forced sparse branch, are *unitary becoming punctuated by rare irreversible commitments* — which is exactly the structure of quantum measurement (Schrödinger evolution interrupted by occasional collapse). That single structure yields:

- **Coherence (quantum mechanics).** Sparse collapse leaves room for unitary evolution between commitments; it avoids the Zeno freeze. Superposition, interference, and coherence exist *because* the universe does not commit every tick. (§5.5)
- **Dilation (gravitational time dilation).** Becoming proceeds at a position-dependent proper-time rate — the lapse. Time dilates because becoming is uneven; a uniform tick could not bend. (§5.4)
- **Silence (preferred-frame safety).** Sparse becoming means a low local event density, hence a tiny local khronon stiffness, hence a preferred-frame footprint suppressed by tens of orders. (§4–§6)

These are not three coincidences. They are one mechanism seen three ways. **Sparse becoming is the hinge on which ED’s quantum and gravitational sectors turn.** The universe does

not collapse every tick — and that single fact explains why it can interfere, dilate, and pass the preferred-frame tests. **The arrow’s alibi is coherence itself.**

That a *correct* structural reading should do this much work at once — that the very property required for quantum mechanics is the property that makes the gravitational theory safe — is, we suggest, evidence that sparse becoming is the right reading of ED, and not a convenient branch chosen after the fact.

9. Relation to GR-II and the Existing Gravity Arc

This paper refines GR-II; it does not contradict it. **GR-II fixed the class; GR-IV fixes the size.** GR-II established the gravitational *class* — that ED is khronometric, Einstein plus one physical scalar — from a gauge-group and mode-count argument. It did not compute the *magnitude* of the khronon’s couplings. This paper supplies that magnitude and finds it density-suppressed: ED is khronometric in form (GR-II), with couplings that are tiny except at Planck density (here). The khronon is exactly as physical as GR-II says; it is simply, in any environment we can test, very weakly coupled. The natural summary is that ED is a *screened* khronometric theory.

The result integrates with the rest of the arc without strain. Newton’s constant $G = c^3 \ell_P^2 / \hbar$ (Paper 027) sets the band-fraction ratio k_{11}/s_{02} that prefactors λ_{local} . The transition acceleration $a_0 = cH_0/(2\pi)$ (Paper 029) is the cosmological face, untouched by the local suppression (§7). The Combination Rule (Paper 030) and slope-4 BTFR (Paper 031) live in the deep-MOND regime governed by a_0 , not λ_{local} . And the cosmological-constant scale (KM-II, Paper 038.5) is a property of the same cosmic boundary R_H . The local face is faint; the cosmological face carries the dark sectors. One field, two scales.

10. Falsification Criteria

The theory can lose, and here is where. (*Each criterion is stated so that a definite measurement could decide it.*)

- **F1 — a measured preferred-frame effect.** If α_1 is measured nonzero at or above $\sim 10^{-5}$ in any environment, the density-suppression mechanism is excluded. (The prediction is that it is unobservably small in all sub-Planck-density tests.) A nonzero α_2 is sharper still: ED predicts $\alpha_2 = 0$ *exactly* (the luminal cones, §3), so *any* detection of α_2 falsifies ED-khronometric gravity at the structural level, not merely the suppression mechanism.
- **F2 — a dense vacuum.** If the ED vacuum event density is shown to be Planck-dense — equivalently, if commitment is shown to fire at every tick — then $\lambda_{\text{local}} = O(1)$ and the theory is excluded. (This is the dense branch, which §5.5 argues abolishes quantum mechanics.)
- **F3 — a broken transition scale.** If the galactic transition acceleration departs from $a_0 = cH_0/(2\pi)$ at high precision (beyond the Hubble-tension band), the cosmological face — which carries MOND independently of λ_{local} — is wrong.
- **F4 — a quantum contradiction.** Because the sparse-becoming structure is the *same* structure as quantum measurement, an experiment that contradicts ED’s identification

of commitment with sparse decoherence — that forces continuous, dense determination — would simultaneously break the α_1 suppression and ED’s quantum sector. The two stand or fall together.

The fourth is the sharpest expression of the paper: in ED, the gravitational preferred-frame safety and the existence of quantum mechanics are not independently adjustable. **The same mechanism that keeps the universe quantum keeps it relativistic.**

11. Position Statement

ED is a screened khronometric theory: Einstein locally, Milgrom cosmologically, distinct only at the Planck frontier. The arrow of time is in the law — it forces the preferred foliation, the khronon, the khronometric class. But its observational footprint is set by the local density of becoming, and becoming is sparse, because dense becoming would freeze the universe. So the preferred-frame parameters that would have killed the theory are suppressed by tens of orders of magnitude, the modified-dynamics scale survives intact on the cosmic boundary, and the theory looks exactly like General Relativity everywhere it has been tested.

The deeper claim is the one the program did not expect to make. **The arrow is present in the law, absent in the data, and essential to both.** Sparse becoming is not a parameter to be tuned — it is the principle that makes ED a theory at all: the same rarity of commitment that lets the universe stay coherent (quantum mechanics) lets its clock bend (time dilation) and keeps its preferred frame quiet (preferred-frame safety). **The arrow is silent where it must be, and loud where it should be.**

We do not claim ED is confirmed. We claim that its sharpest falsification front — the one the One-Field letter flagged as a kill-switch — does not kill it, and does not for a reason internal to what ED is. The theory remains conditional on its primitives, falsifiable at the marked fronts, and consistent with observation where it has been pushed. But on the preferred-frame question, the arrow has an alibi, and the alibi is coherence itself.

Appendix

A. Derivation details for λ_{local}

The metric (gravitational) stiffness M_P^2 is the coefficient of the spatial-curvature energy and originates in P02 reciprocal sharing — present at every adjacency every tick, hence $M_P^2 \sim s_{02}/\ell_P^2$, density-independent. The khronon stiffness f^2 is the coefficient of the foliation-expansion energy $(\nabla \cdot u)^2$ and originates in P11 commitment-pinning — present only where commitments fire, hence $f^2 \sim k_{11} \rho_{\text{event}} \ell_P^2$. The dimensionless coupling is the ratio $\lambda_{\text{local}} = f^2/M_P^2 = (k_{11}/s_{02}) \rho_{\text{event}}/\rho_{\text{Planck}}$, with $k_{11}/s_{02} = O(1)$ fixed by $\kappa/D = 8\pi G$ (GR-III). Linear in the dilute regime ($\lambda \propto \rho_{\text{event}}$), saturating toward $O(1)$ as $\rho_{\text{event}} \rightarrow \rho_{\text{Planck}}$.

B. The event-density hierarchy

ρ_{event} is the local density of irreversible determination events per four-volume; $\rho_{\text{Planck}} = \ell_P^{-4} \approx 5 \times 10^{113} \text{ J/m}^3$ in energy-density units. Estimates for the density the khronon samples span

the Solar-System field-energy scale ($\sim 10^{-110} \rho_{\text{Planck}}$), the source matter scale ($\sim 10^{-93}$), the cosmological floor ($\sim 10^{-122}$), and a nuclear-density stress-test ($\sim 10^{-78}$). All are deeply sub-Planck; the verdict is robust to the choice.

C. The Zeno-limit argument

A continuously measured quantum system is confined to its instantaneous eigenstate (the quantum Zeno effect): the survival probability over a time t subdivided into n measurements tends to unity as $n \rightarrow \infty$, freezing transitions. Identifying commitment (P11, indeterminate $\rightarrow$ determinate) with measurement, dense commitment (a measurement every tick) is the $n \rightarrow \infty$ limit — frozen evolution, no superposition, no dynamics. ED's quantum sector therefore forbids dense commitment; commitment is sparse.

D. Summary of Paper 029's dipole projection

The cosmic decoupling surface $R_H = c/H_0$ supplies anisotropic content $\rho_{\text{cosmic}}(\theta) \sim \rho_0(|\vec{a}|/c) \cos \theta$ with $\rho_0 = cH_0$. An accelerating chain's $SO(3) \rightarrow SO(2)$ reduction makes the dipole ($\ell = 1, m = 0$) the leading mode; the azimuthal-Fourier normalization on the residual $SO(2)$ (period 2π) contributes the factor $1/(2\pi)$, and the polar integration $\int_0^\pi \sin \theta \cos^2 \theta d\theta = 2/3$ together with the substrate-source normalization give $a_{\text{induced}} = cH_0/(2\pi) = a_0$. No local khronon coupling enters.

GR-IV. Closes the preferred-frame falsification front of the ED gravity line. $\alpha_2 = 0$ exactly (both cones luminal, $c_T = c$, $c_s = c$; literature-verified) so the tighter bound is met structurally; the remaining PPN preferred-frame parameter reduces to the local khronon stiffness, $\alpha_1 = -4\lambda_{\text{local}}$ (from $c_T = c$, $c_s = c$); the stiffness is the commitment sparsity, $\lambda_{\text{local}} = (k_{11}/s_{02}) \rho_{\text{event}}/\rho_{\text{Planck}}$, because the metric stiffness is always-on P02 sharing while the khronon stiffness is sparse P11 commitment; and commitment is forced sparse because dense commitment is the quantum Zeno limit (no quantum mechanics). Hence $\alpha_1 \lesssim 4 \times 10^{-93}$ — safe by $\gtrsim 70$ orders. MOND ($a_0 = cH_0/2\pi$, Paper 029) is the khronon's cosmological face and is untouched. Sparse becoming gives ED quantum coherence, gravitational time dilation, and preferred-frame safety — one structure, three faces. ED is a screened khronometric theory: observationally General Relativity, distinct only at the Planck-density frontier. Conditional on its primitives, falsifiable at four marked fronts; refines GR-II (form) with the coupling magnitude; no number fabricated.